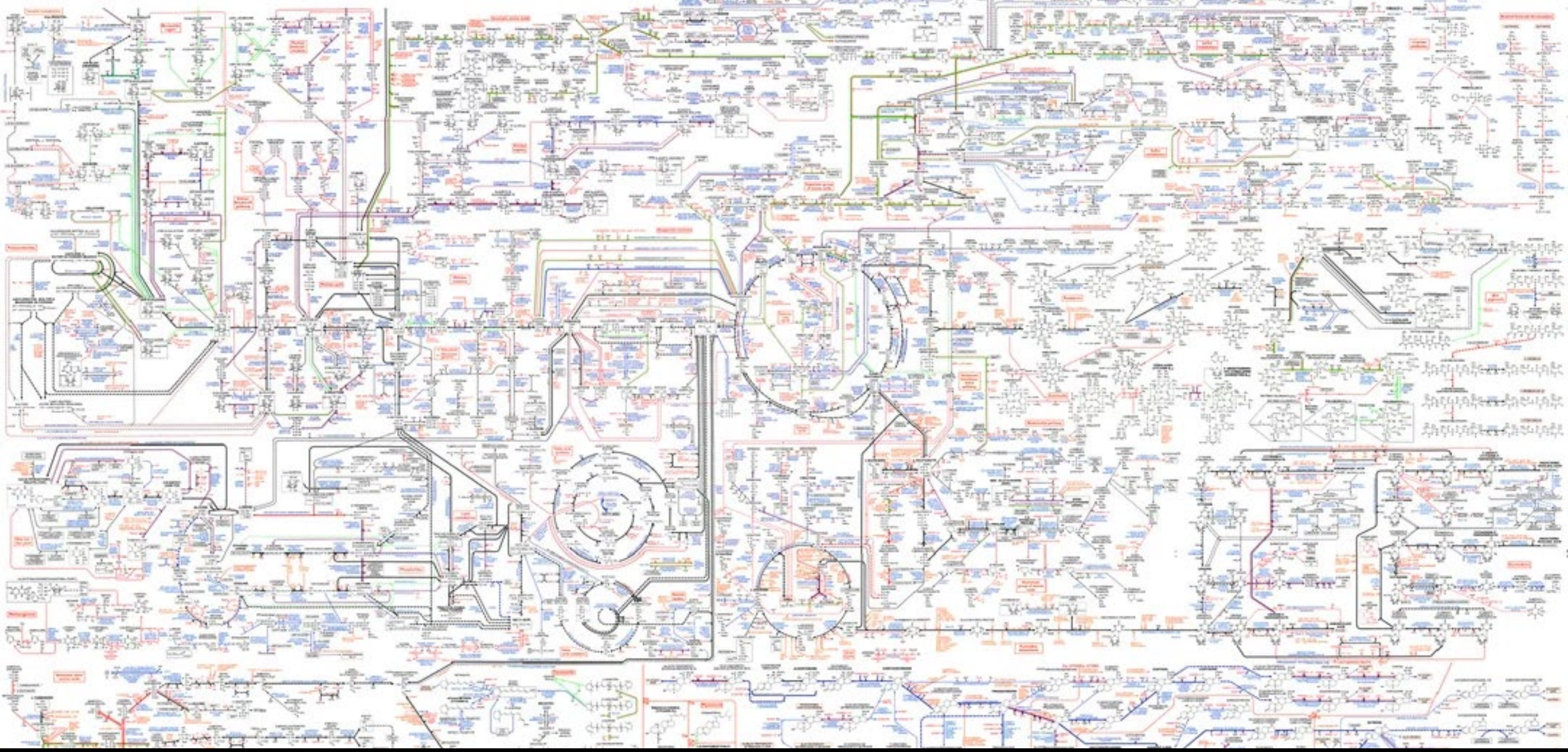
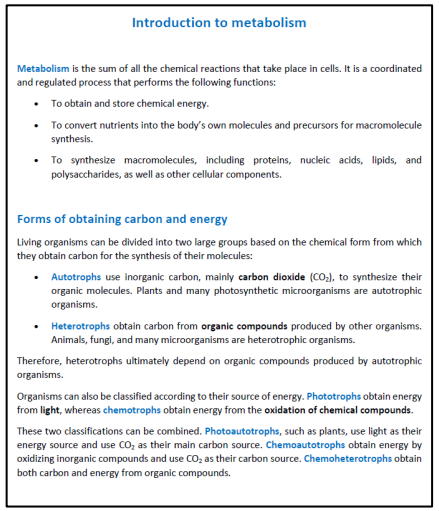


Introduction to metabolism

Pathways, regulation and ATP

Supportive material



Introduction to metabolism

Metabolism is the sum of all the chemical reactions that take place in cells. It is a coordinated and regulated process that performs the following functions:

- To obtain and store chemical energy.
- To convert nutrients into the body's own molecules and precursors for macromolecule synthesis.
- To synthesize macromolecules, including proteins, nucleic acids, lipids, and polysaccharides, as well as other cellular components.

Forms of obtaining carbon and energy

Living organisms can be divided into two large groups based on the chemical form from which they obtain carbon for the synthesis of their molecules:

- Autotrophs use inorganic carbon, mainly **carbon dioxide** (CO_2), to synthesize their organic molecules. Plants and many photosynthetic microorganisms are autotrophic organisms.
- Heterotrophs obtain carbon from **organic compounds** produced by other organisms. Animals, fungi, and many microorganisms are heterotrophic organisms.

Therefore, heterotrophs ultimately depend on organic compounds produced by autotrophic organisms.

Organisms can also be classified according to their source of energy. Phototrophs obtain energy from **light**, whereas chemotrophs obtain energy from the **oxidation of chemical compounds**.

These two classifications can be combined. Photoautotrophs, such as plants, use light as their energy source and use CO_2 as their main carbon source. Chemoautotrophs obtain energy by oxidizing inorganic compounds and use CO_2 as their carbon source. Chemoheterotrophs obtain both carbon and energy from organic compounds.

Additional study material

ADI document: *Introduction to metabolism*

Learning goals

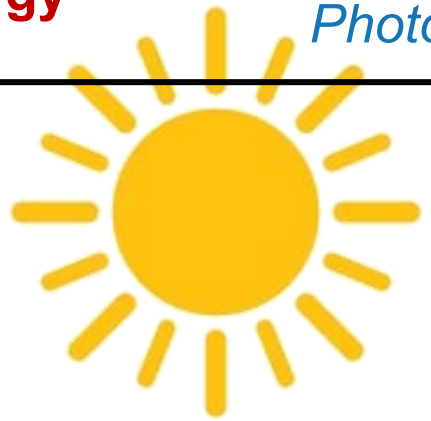
- Basic concepts of metabolism and metabolic pathways
- Types of reactions in metabolism
- Regulatory mechanisms in metabolism
- ATP structure and function

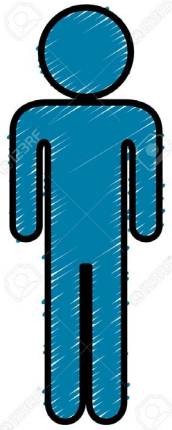
Life uses energy

- Living organisms obtain, store and use energy
- Energy cannot be created or destroyed, only transformed
- Living organisms obtain usable energy from their surroundings and release part of it as heat

Forms of obtaining carbon and energy

Source of carbon	Inorganic (mainly CO_2) *Autotrophs*	Organic *Heterotrophs*
Source of energy	Light *Phototrophs*	Chemical compounds *Chemotrophs*

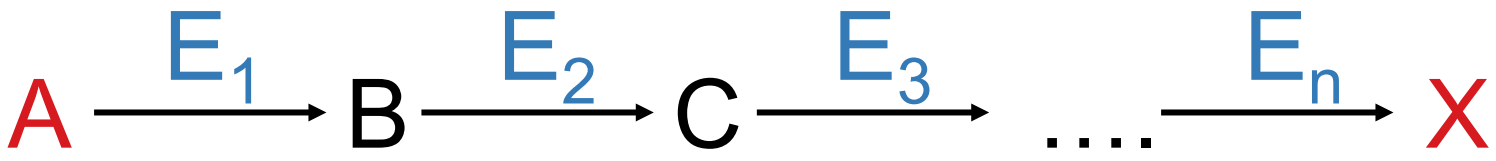




Humans are chemoheterotrophs

Metabolic pathway

A metabolic pathway is a sequence of enzyme-catalyzed reactions that transforms one or more initial substrates into final products through a series of small chemical changes



The intermediate compounds formed along the pathway are called metabolites

Types of chemical reactions in metabolism

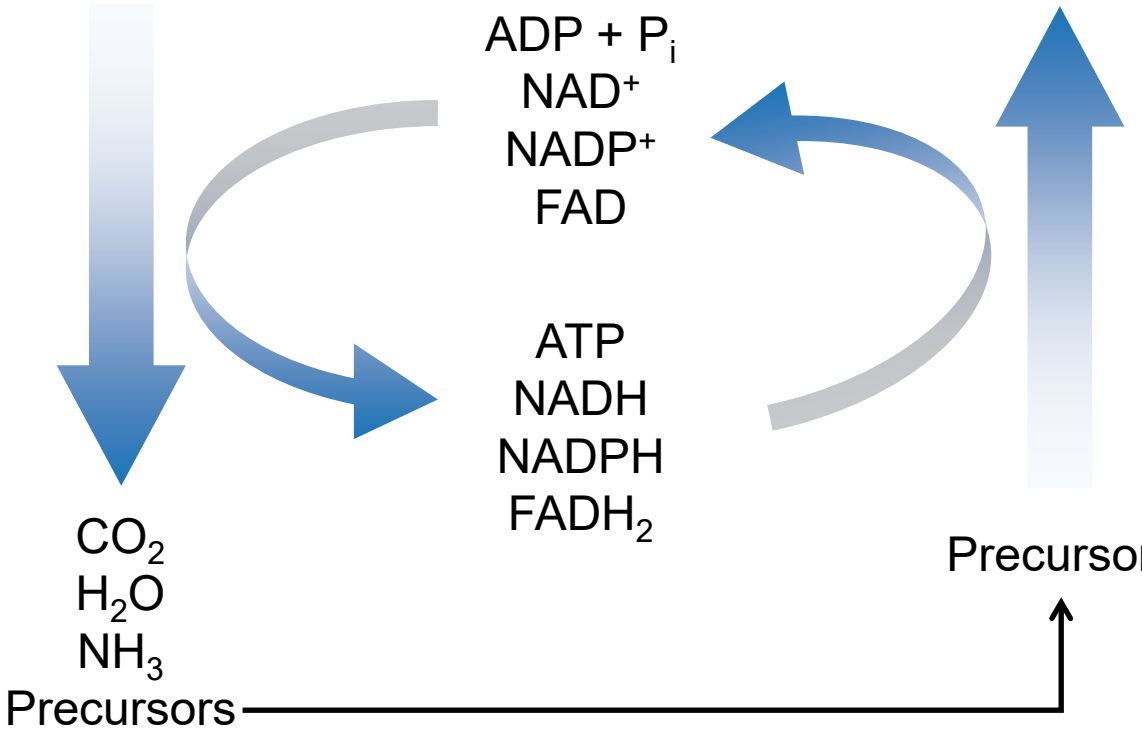
1. Transfer of functional groups
2. Oxidation-reduction reactions
3. Isomerization
4. Hydrolysis and condensation
5. Formation and cleavage of carbon-carbon bonds

Metabolic phases

CATABOLISM

Energy-containing nutrients
Carbohydrates
Lipids
Proteins

Catabolism releases energy and reducing equivalents



ANABOLISM

Macromolecules
Carbohydrates
Lipids
Proteins
Nucleic acids

Anabolism requires energy and reducing power

$$Glucose\ (C_6H_{12}O_6) + 6\,O_2 \rightarrow 6\,CO_2 + 6\,H_2O + energy$$





Catabolism



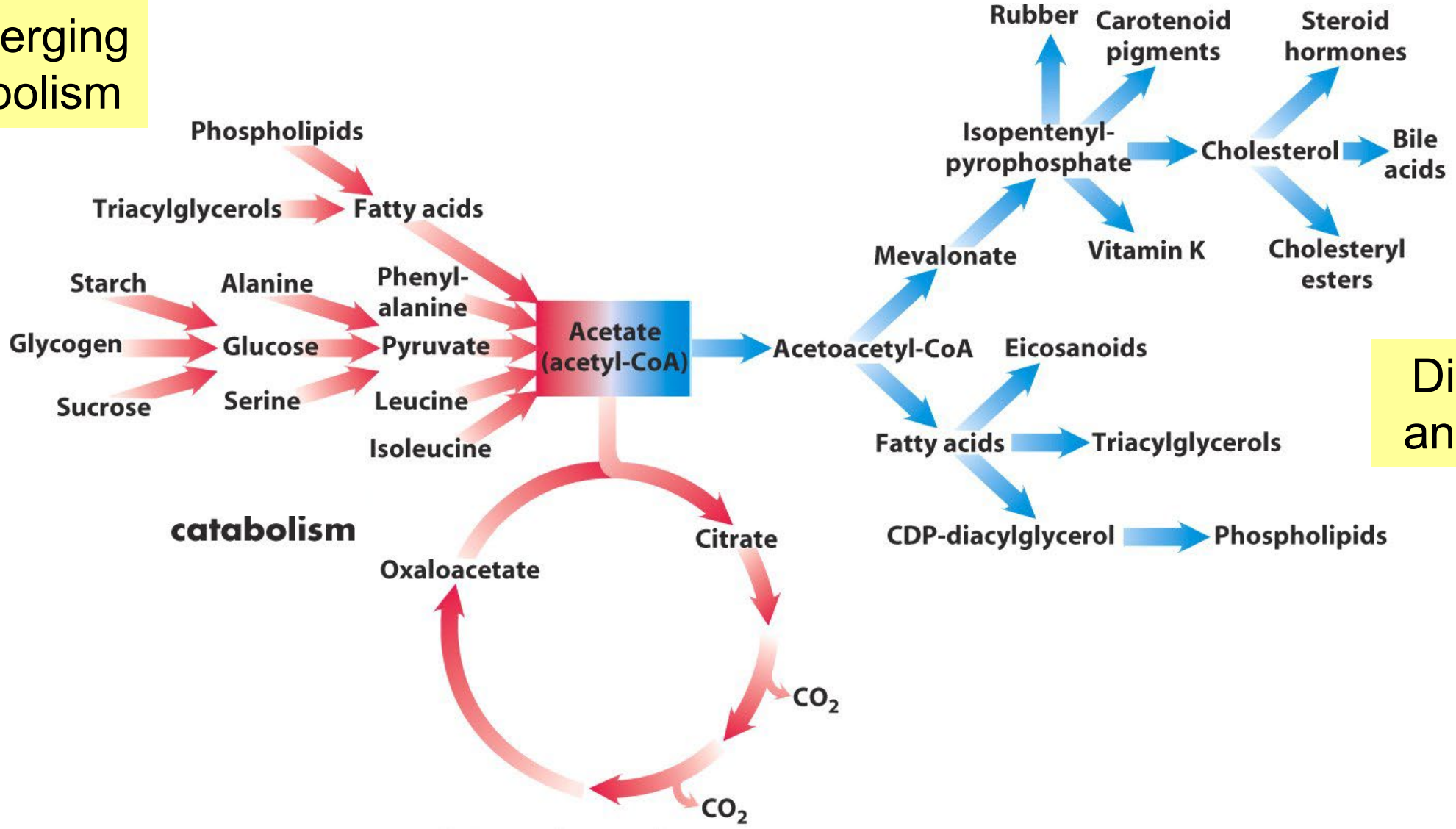
ENERGY

Anabolism



Types of metabolic pathways

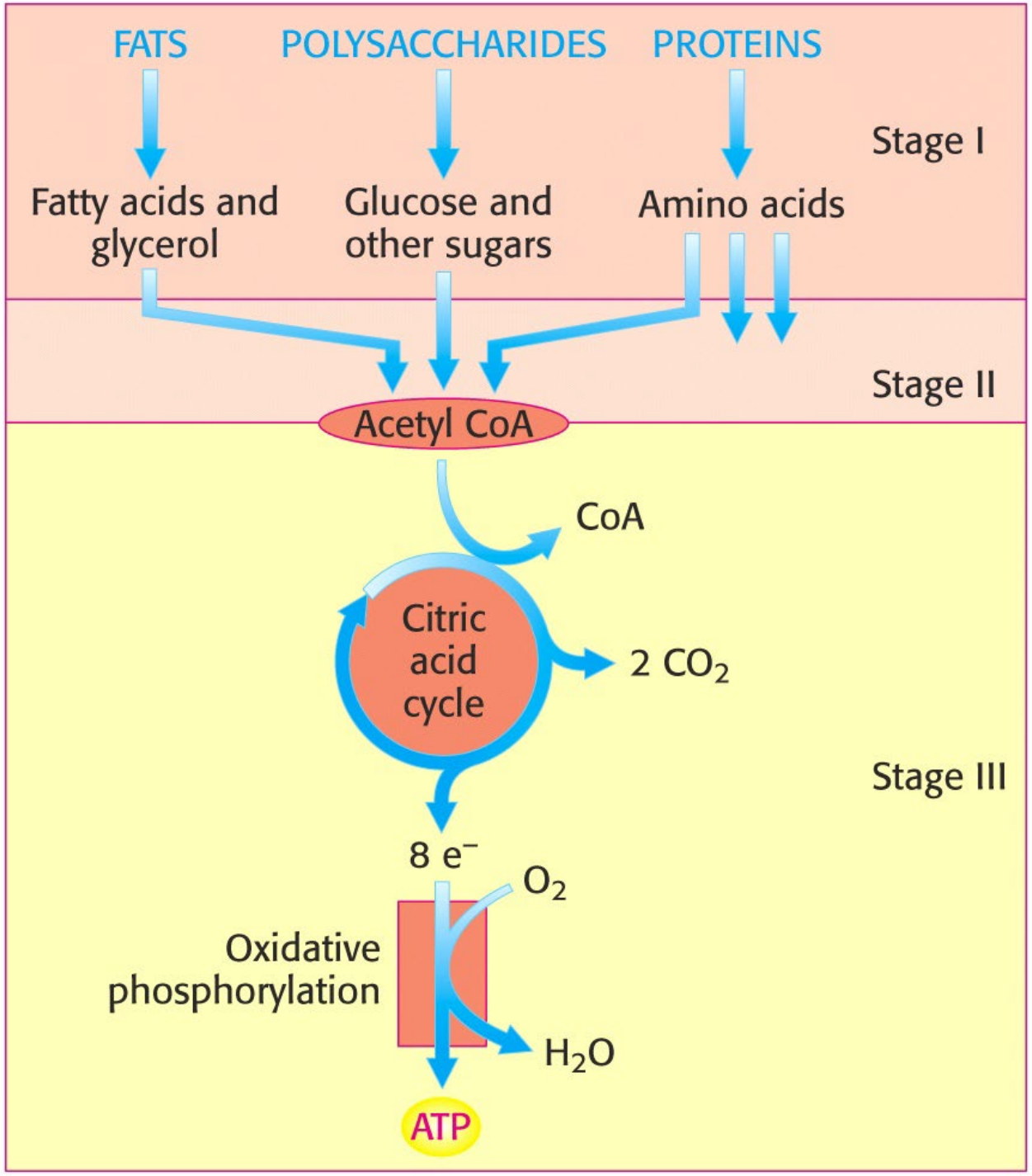
Converging catabolism



Diverging anabolism

Lehninger, 4th ed.

Catabolic stages



Mathews, 3rd ed.

Regulatory mechanisms in metabolism

1. Control of enzyme abundance

   Balance between enzyme synthesis and degradation

2. Control of enzyme activity
   - Allosteric regulation
   - Covalent modification

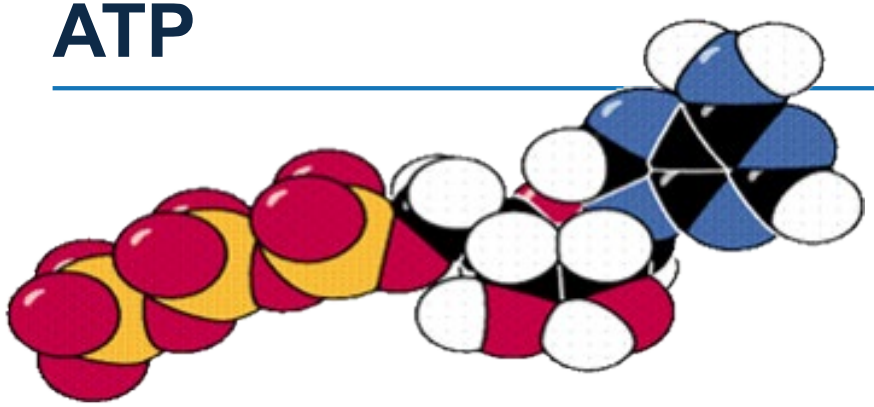
3. Compartmentalization of enzymes and substrates

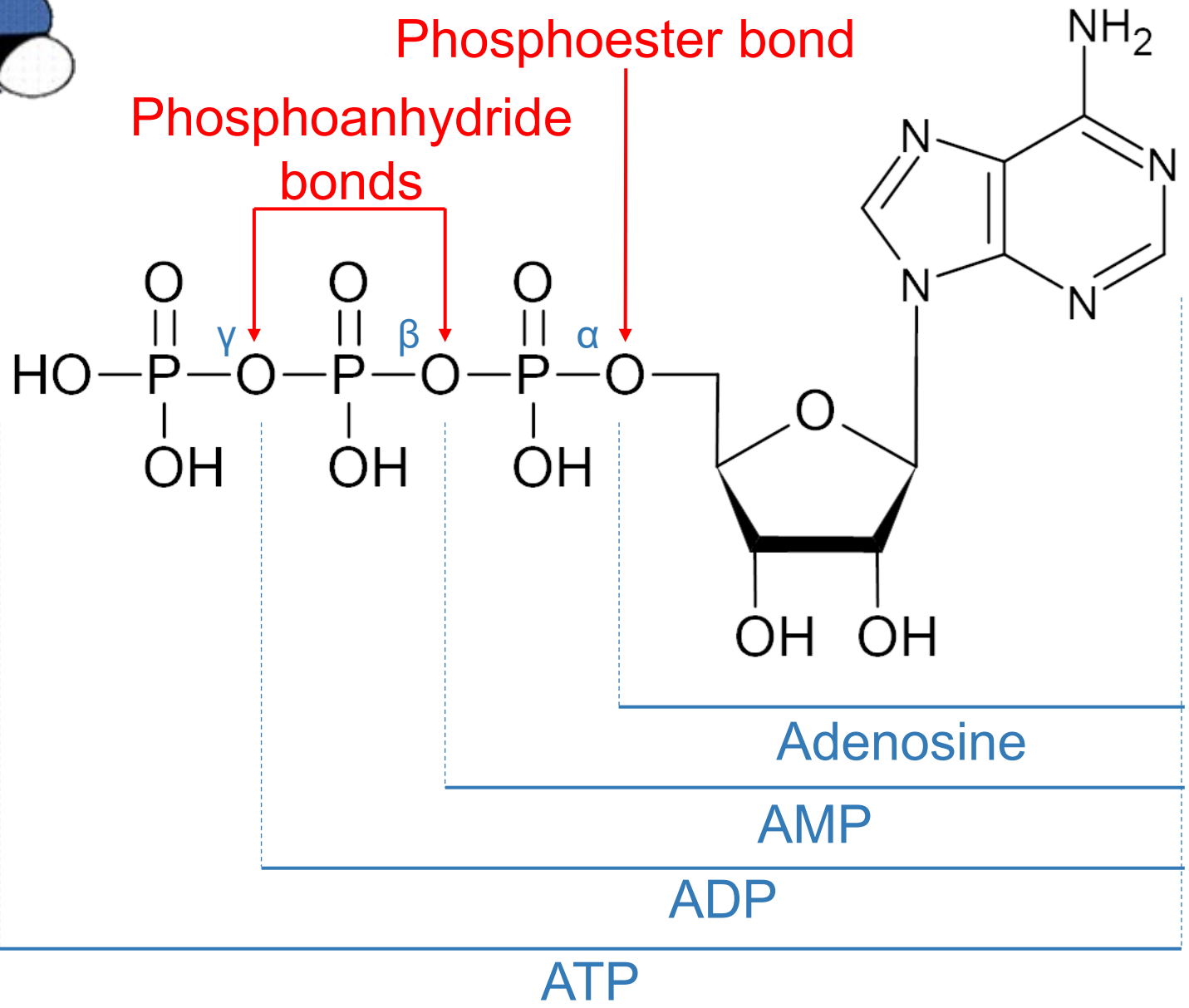
Coordinated by hormonal and neural regulation

Energy control in metabolism

- Cells use and store nutrients according to their energy requirements and the availability of metabolic resources
- Cells use energy for:
  - Mechanical work
  - Active transport across membranes
  - Synthesis and maintenance of structures (biomolecules)
- The cellular energy state, reflected in the relative concentrations of ATP, ADP and AMP, is a major factor in metabolic regulation
- ATP is the main immediate energy carrier in cellular metabolism

ATP




Phosphoester bond
Phosphoanhydride bonds
NH2
N
N
N
N
O
γ
β
α
HO—P—O—P—O—P—O
OH
OH
OH
O
OH OH
Adenosine
AMP
ADP
ATP

ATP hydrolysis

$$ATP + H_2O \rightarrow ADP + P_i \qquad \Delta G^{o\prime} = -30.5 \text{ kJ/mol}$$